Improving Hallucination Detection via Question-Aware Knowledge Summarization

***Abstract*—Hallucinations — fluent answers that are not supported by the provided evidence — remain a central reliability concern in knowledge-grounded question answering, where every answer is expected to be justified by an explicit knowledge passage. We hypothesize that a substantial share of detection errors arises not from the verifier alone but from irrelevant context that obscures the evidence needed to judge an answer, and we ask whether improving the signal-to-noise ratio of the passage is as valuable as improving the verifier itself. To examine this, we pair a deliberately simple supervised verifier with question-aware knowledge summarization. The verifier is a fine-tuned DeBERTa-v3-base model that operates directly on evidence–question–answer (K, Q, A) triples, requiring no retrieval ensembles, sampling, or access to model internals, and producing a deterministic prediction for each input. As a preprocessing step, each passage is summarized conditioned on the question so that question-relevant evidence is retained while extraneous content is removed before verification. Across three summarization backbones (BART, FLAN-T5, and PEGASUS), summarization consistently improves detection while shortening inputs and reducing inference cost. On HaluEval-QA, our best configuration reaches an F1 of 0.9849. The effect is most pronounced under distribution shift: on an unseen HotpotQA-derived benchmark, where passages are long and reasoning-intensive, question-aware summarization raises macro-F1 from near-chance (0.4662) to 0.7905 — precisely the setting in which irrelevant context is heaviest. Taken together, these results indicate that reducing irrelevant context is a key lever for reliable hallucination detection, and that a lightweight supervised verifier combined with targeted summarization offers a deterministic, low-overhead solution well suited to black-box QA systems.**

***Index Terms*—Hallucination Detection, Knowledge-Grounded QA, DeBERTa, Supervised Learning, Question-Aware Knowledge Summarization, HaluEval**

I. Introduction

Large language models (LLMs) can generate fluent and coherent answers that are not supported by the provided evidence. In knowledge-grounded question answering, reliable reasoning depends on the quality and representation of the underlying knowledge provided to the model. Prior work has shown that extracting and organizing information from data is critical for improving knowledge utilization and decision making [1]. This phenomenon, commonly referred to as hallucination, poses a serious risk in knowledge-grounded question answering (QA), where the user expects the answer to be justified by an explicit knowledge passage rather than the model's internal parametric memory.

We hypothesize that reliable hallucination verification depends primarily on accurately modeling the relationship between the provided evidence and the candidate answer, rather than relying on internal generation signals. As a result, verification should focus on evidence–answer consistency under the given context.

A practical mitigation is a verification layer that evaluates whether a generated answer is grounded in the provided knowledge. Formally, given a knowledge passage K, a question Q, and a candidate answer A, a verifier predicts whether A is supported by K (non-hallucination) or not (hallucination). Such a verifier can be integrated into QA systems to filter hallucinated outputs, trigger regeneration, or abstain when confidence is low.

In practice, knowledge passages can be long and may include information that is irrelevant to the question. This irrelevant context can introduce noise and make evidence verification more difficult. Reducing such noise is therefore important for effective hallucination detection.We argue that a major source of hallucination detection errors is not only the verification model itself, but also the presence of irrelevant context that obscures the evidence required for verification. Therefore, improving the signal-to-noise ratio of the provided knowledge may be as important as improving the verifier architecture itself.

Existing hallucination detection approaches differ in assumptions and operational cost. White-box methods rely on internal model signals and require privileged access. Ensemble and calibration-based frameworks may improve accuracy but introduce significant system complexity and higher inference cost. Judge-based LLM evaluation is flexible but expensive and not optimized for deterministic binary verification. These limitations motivate a simple, efficient, and reproducible supervised solution that operates purely on textual inputs.

A. Contributions

The main contributions of this work are:

- We demonstrate that question-aware knowledge summarization removes irrelevant context and improves evidence-answer alignment for hallucination detection.
- We systematically show that question-aware evidence compression improves verification accuracy, computational efficiency, and external generalization on long-context benchmarks.
- We develop a lightweight supervised verifier that enables controlled evaluation of evidence compression under black-box deployment constraints.
- We analyze question-unaware knowledge summarization and show that evidence compression alone can remain competitive, clarifying the role of summarization independent of explicit question conditioning.
- We present a structured comparison of hallucination detection approaches in terms of performance, complexity, and deployability.

The remainder of this paper is organized as follows. Section II reviews related work on hallucination detection. Section III describes the dataset and problem setup. Section IV presents the proposed verification framework. Section V introduces the knowledge summarization component. Section VI reports the experimental setup and results. Section VII discusses practical deployment considerations. Finally, Section VIII concludes the paper and outlines future research directions.

II. Related Work

This section reviews prior work on hallucination detection in knowledge-grounded QA and highlights key limitations that motivate our approach.

A. Hallucination Detection Benchmarks

HaluEval provides a controlled benchmark for hallucination analysis across multiple tasks, including QA [2]. Each QA instance includes a knowledge passage, a question, and both grounded and hallucinated answers, enabling supervised learning of detectors. Recent benchmarks emphasize controlled hallucination generation to support systematic evaluation. However, HaluEval does not prescribe a specific detection model, leaving room for model design choices and practical deployment strategies.

B. White-box Hallucination Detection

White-box methods exploit internal model representations and gradients. EGH is a representative approach [3] that extracts embeddings and gradient-based features to predict hallucination. Several recent studies follow similar directions by leveraging internal confidence or inconsistency signals. Although effective, white-box assumptions require access to model internals. This requirement limits applicability in black-box or API-based systems and complicates real-world deployment.

C. Ensemble and Calibrated Verification Frameworks

Ensemble-based frameworks such as HALT-RAG combine multiple verification signals and calibration strategies [4]. Recent retrieval-augmented detection frameworks [5] integrate external knowledge and multi-stage reasoning to improve factuality assessment. While these systems can achieve strong performance, they introduce additional components such as retrieval modules, calibration mechanisms, or sampling strategies. This increases inference cost, engineering complexity, and operational overhead.

D. Judge-based Hallucination Evaluation

Judge-based approaches treat hallucination detection as an evaluation problem. LLM-as-a-Judge methods prompt large language models to score answers [6]. Sampling-based frameworks such as SelfCheckGPT [7] assess consistency across multiple generations. Recent work such as MetaQA [8] improves upon sampling-based approaches by introducing metamorphic relations and response mutation for more robust hallucination detection without external resources. While flexible and model-agnostic, these approaches rely on stochastic sampling or prompt sensitivity. They are computationally expensive at scale and are not optimized for deterministic, task-specific binary verification.

E. Lightweight Feature-based Hallucination Detectors

Recent work such as [9] investigates lightweight hallucination detectors based on surface lexical features and classical machine learning classifiers. These methods typically represent the input text using sparse TF-IDF features and combine them with simple numeric signals such as lexical overlap between the generated answer and the supporting context, answer length, or numerical consistency. Such feature-based detectors are computationally efficient and easy to reproduce, making them useful baselines for large-scale hallucination analysis [9]. However, because they rely primarily on lexical cues and handcrafted features, their ability to capture deeper semantic relationships between evidence and generated answers is limited. As a result, these approaches may struggle to detect hallucinations that require contextual reasoning rather than surface-level pattern matching.

F. Reference-based Factual Consistency Evaluation

Recent work has proposed reference-based factual consistency metrics that evaluate whether generated content is supported by a source document or evidence passage.

AlignScore [10] measures factual consistency through semantic alignment between a source text and generated content. It employs a unified alignment function trained on multiple language understanding tasks, enabling robust evaluation across diverse factual consistency scenarios. Unlike retrieval-based or sampling-based approaches, AlignScore directly estimates the degree of support between evidence and generated text, making it suitable for black-box evaluation settings.

MiniCheck [11] is a lightweight factual consistency verification framework that determines whether a generated claim or answer is supported by the provided context. Unlike computationally expensive ensemble-based approaches, it performs document-grounded fact verification efficiently while maintaining competitive performance. Its low computational cost and strong factuality assessment capabilities make it a practical baseline for hallucination detection.

AlignScore and MiniCheck are closely related to knowledge-grounded hallucination detection because they assess evidence-answer consistency without requiring access to internal model representations.

Overall, existing approaches involve clear tradeoffs between performance, complexity, and deployability. White-box methods rely on privileged access to internal representations. Retrieval-augmented and ensemble systems depend on multi-stage pipelines and external knowledge integration. Judge-based methods rely on sampling or prompt design and may lack deterministic behavior.

In contrast, relatively few lightweight black-box approaches formulate hallucination detection as a direct supervised classification task over (K, Q, A) triples. Such a formulation enables end-to-end optimization, deterministic inference, and simplified deployment under black-box constraints.

Our experimental results demonstrate the effectiveness of this formulation. A single fine-tuned DeBERTa-v3 verifier operating directly on (K,Q,A) triples achieves strong performance on HaluEval-QA and competitive generalization on external benchmarks. On HaluEval-QA, it reaches an F1 score of 0.9773 using original knowledge and improves to 0.9849 when combined with question-aware summarization (Table VII). Importantly, this performance is obtained without relying on retrieval modules, ensemble methods, or multi-stage pipelines, indicating that end-to-end optimization over textual inputs is sufficient for high-quality hallucination detection.

In addition, the proposed verifier performs deterministic inference, where each (K, Q, A) input produces a single consistent prediction without reliance on sampling or stochastic generation. This contrasts with judge-based or sampling-based approaches and ensures reproducibility and stable behavior across runs. Furthermore, the model operates purely on textual inputs and does not require access to internal model representations, making it suitable for deployment under black-box constraints with reduced system complexity.

Furthermore, long knowledge passages may introduce irrelevant or noisy information that can degrade hallucination detection performance. Our results confirm that reducing such noise through question-aware knowledge summarization consistently improves both verification accuracy and F1 score while reducing input length, as shown in Table VII. This demonstrates that improving the signal-to-noise ratio enhances evidence–answer alignment while also improving computational efficiency.

Overall, this work targets that gap by proposing a single fine-tuned verifier and further improving robustness through question-aware knowledge summarization.

Unlike prior work that focuses on architectural modifications, retrieval augmentation, or ensemble verification, the contribution of this work is a lightweight end-to-end hallucination detection framework that combines supervised evidence-answer verification with question-aware knowledge summarization.

The framework operates entirely under black-box constraints, requires no retrieval module, calibration stage, or model-internal signals, and achieves strong performance while maintaining low deployment complexity and deterministic inference.

TABLE I: Comparison of hallucination detection approach families in terms of methodology and deployment practicality.

Approach	Supervised	Fine-tuned	Retrieval	Model Access	Deterministic	Complexity	Deployability
Feature-based Detector [9]	Yes	No	No	Black-box	Yes	Low	High
EGH [3]	Yes	Partial	No	Internal	Yes	Medium	Low
HALT-RAG [4]	Hybrid	No	Yes	Black-box	No	High	Medium
LLM-as-a-Judge [6]	No	No	Optional	Black-box	No	High	Low
SelfCheckGPT [7]	No	No	No	Black-box	No	Medium	Medium
Ours	Yes	Yes	Optional	Black-box	Yes	Low	High
Ours + Summarization	Yes	Yes	Optional	Black-box	Yes	Low	Very High

Complexity definition: Complexity denotes system and deployment overhead, considering model access requirements, number of components, inference cost, and dependence on external retrieval or sampling.

III. Dataset and Problem Setup

A. HaluEval-QA

HaluEval-QA pairs each question with a knowledge passage and includes both grounded and hallucinated answers [2]. Each original record is converted into two supervised samples: (K, Q, A_{true}) labeled as non-hallucination (1) and (K, Q, A_{hall}) labeled as hallucination (0). This construction yields a balanced binary classification dataset aligned with our verification objective.

While this controlled construction enables reliable and fair evaluation, it may introduce biases related to dataset balance and the specific manner in which hallucinated answers are generated. As a result, performance may differ in real-world settings with unbalanced data, noisier retrieval, or adversarial hallucinations.

B. Data Split

We use a 70/30 train-test split at the original-record level before constructing the final binary verification samples. The original HaluEval-QA records are first partitioned into training and test sets, after which each record is converted into two verification instances: a supported-answer sample (label = 1) and a hallucinated-answer sample (label = 0). The resulting test set contains 6,000 balanced samples, ensuring comparable class support when reporting accuracy and F1.

To prevent data leakage, the train-test split was performed before constructing the final binary verification samples. Specifically, the original HaluEval-QA records were first partitioned at the knowledge-passage level and then converted into supported and hallucinated verification instances within the same split. Consequently, the supported and hallucinated variants derived from the same original record never appear across both training and test sets.

To further verify the integrity of the experimental setup, we performed an explicit leakage analysis between the resulting splits. The analysis confirmed zero question overlap, zero knowledge-passage overlap, zero shared original records, and zero sibling leakage between the training and test sets.

TABLE II: Dataset statistics used in our experiments.

Split	#Instances	#Hallucination (0)	#Non-hallucination (1)
Train	14,000	7,000	7,000
Test	6,000	3,000	3,000

These checks ensure that the reported performance reflects generalization to previously unseen questions and evidence passages rather than memorization of duplicated or related samples.

Example 1 (Supported Answer, Label = 1)

Question: What anime did Unscandal perform theme music for and is written and illustrated by Jin Kobayashi?

Answer: School Rumble

Knowledge (excerpt): The band is most well known for performing the theme music for the anime School Rumble. School Rumble is a Japanese manga series written and illustrated by Jin Kobayashi.

Interpretation: The answer is directly supported by the evidence and is therefore labeled as non-hallucinated.

Example 2 (Hallucinated Answer, Label = 0)
Question: Joey Richter co-starred as which Harry Potter character in 2009 fan parody musicals?
Answer: Joey Richter co-starred as Harry.
Knowledge (excerpt): Richter co-starred as Ron Weasley in the fan-parody musicals ‘A Very Potter Musical” (2009), ‘A Very Potter Sequel” (2010), and “A Very Potter Senior Year” (2012). Ron Weasley is a fictional character in J. K. Rowling’s Harry Potter series.
Interpretation: The answer contradicts the evidence because the knowledge passage identifies the character as Ron Weasley rather than Harry. Therefore, the answer is labeled as hallucinated.

The examples demonstrate the two target classes used in our verification task. Supported answers are grounded in the provided evidence, whereas hallucinated answers introduce information that is inconsistent with the knowledge passage.

IV. Method

Hallucination verification relies on assessing the consistency between a candidate answer and the provided evidence. In practice, knowledge passages may be long and contain irrelevant information. Such noise can make verification more difficult. This motivates approaches that reduce irrelevant context while preserving a simple verification pipeline.

Figure 1 illustrates the overall architecture of the proposed framework.

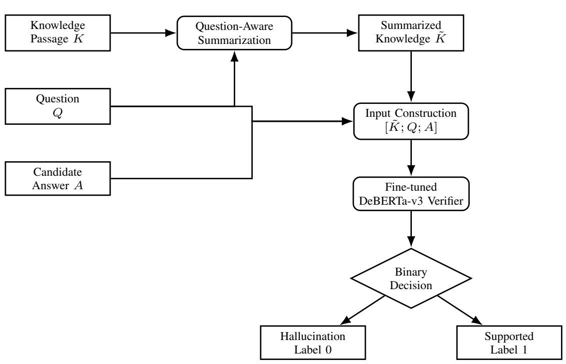


Fig. 1: Overview of the proposed framework. Question-aware summarization reduces context noise and preserves question-relevant evidence, improving hallucination detection on long knowledge passages. The summarized evidence is then verified using a lightweight supervised classifier.

A. Task Definition

Given a knowledge passage K, a question Q, and a candidate answer A, our goal is to predict a binary label $y \in \{0, 1\}$ indicating whether A is supported by K (1) or hallucinated (0). The verifier can be used as a post-generation filter in QA pipelines.

B. Representative Dataset Samples

To illustrate the binary verification task, the following examples show one supported answer and one hallucinated answer from the HaluEval-QA dataset.

C. Input Representation

We convert each triple (K, Q, A) into a single sequence for encoder-only verification:

$$\texttt{[CLS]}\ K\ \texttt{[SEP]}\ Q\ \texttt{[SEP]}\ A\ \texttt{[SEP]}$$

This format allows the model to jointly attend to evidence, question, and answer. We truncate or pad inputs to a fixed maximum length of 512 tokens.

D. Model Architecture

We fine-tune `DeBERTa-v3-base` as a lightweight verifier. Compared to earlier encoder models such as RoBERTa or ELECTRA, DeBERTa-v3 employs disentangled attention and enhanced position encoding. These properties improve modeling of semantic relationships under limited context. Let $h_{\texttt{[CLS]}}$ denote the final hidden representation of the `[CLS]` token. A linear classification head produces logits:

$$z = W h_{\texttt{[CLS]}} + b$$

and class probabilities:

$$p(y \mid K, Q, A) = \mathrm{softmax}(z)$$

E. Training Objective

We optimize the cross-entropy loss over the binary labels:

$$\mathcal{L} = - \sum_{c \in \{0,1\}} \mathbb{1}[y = c] \log p(y = c \mid K, Q, A)$$

This objective trains the verifier to separate grounded from hallucinated answers.

F. Question-Unaware Evidence Compression

We additionally study evidence compression without conditioning on the question. The knowledge passage K is summarized based only on its content, producing a compressed version $\hat{K}$. The verifier architecture and training objective remain unchanged. We replace K with $\hat{K}$ during verification to analyze the effect of evidence compression alone.

V. Knowledge Summarization for Detection

Knowledge passages are often long, and much of what they contain has little to do with the question at hand. Pulling out the parts that actually matter — and setting the rest aside — is an old problem in knowledge-based systems [1], and it turns out to matter here too: the cleaner the evidence the verifier sees, the easier its job becomes. This is what motivates question-aware summarization, and in the rest of this section we ask whether trimming that noise genuinely helps hallucination detection.

A. Question-Aware Summarization

We generate a condensed knowledge passage $\tilde{K}$ conditioned on the question Q using pretrained summarization models. Specifically, we prompt the model as: *"Summarize the following knowledge to answer the question. Question: Q. Knowledge: K."* The resulting summary focuses on question-relevant evidence. While summarization can improve the signal-to-noise ratio, it may also omit subtle details or introduce errors, making its impact an empirical question rather than an assumption. We evaluate multiple summarization backbones (BART, FLAN-T5, PEGASUS).

B. Integration with the Verifier

The verifier architecture remains unchanged. We replace the original knowledge K with the summarized knowledge $\tilde{K}$:

$$(K, Q, A) \Rightarrow (\tilde{K}, Q, A)$$

This design isolates the effect of summarization without modifying the classifier.

VI. Experimental Setup and Results

We evaluate two primary settings: (1) using original knowledge passages (K) and (2) using summarized knowledge passages ($\tilde{K}$). All verification hyperparameters are kept fixed across experiments to ensure a fair comparison.

The detector is trained using the HuggingFace Transformers pipeline with AdamW optimization. We report accuracy, precision, recall, and F1 score on a held-out test set of 6,000 samples.

A. Evaluation Metrics

We evaluate the verifier using four standard classification metrics: accuracy, precision, recall, and F1 score. In our binary verification setting, the supported-answer class (label = 1) is treated as the positive class. These metrics are computed from the confusion matrix, where TP denotes true positives, TN denotes true negatives, FP denotes false positives, and FN denotes false negatives. In our binary setting, the supported answer class is treated as the positive class.

Accuracy measures the proportion of correctly classified samples:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision measures how many predicted supported answers are actually supported:

$$Precision = \frac{TP}{TP + FP}$$

Recall measures how many truly supported answers are correctly detected:

$$Recall = \frac{TP}{TP + FN}$$

The F1 score is the harmonic mean of precision and recall:

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

These metrics provide a balanced evaluation of the verifier by measuring both overall correctness and class-specific detection quality.

B. Baseline Methods

To provide a broader comparison with recent factual consistency and hallucination detection approaches, we evaluate two additional black-box baselines on the same 6,000-sample HaluEval-QA test set used throughout this work.

We also include a zero-shot NLI baseline to quantify the gain obtained from supervised fine-tuning compared with an unoptimized pretrained verifier.

AlignScore is a reference-based factual consistency metric that measures semantic alignment between the knowledge passage and the candidate answer. Predictions are obtained through threshold-based classification of the alignment score.

MiniCheck is a lightweight factual consistency verification framework that predicts whether a candidate answer is supported by the provided context.

Both baselines are evaluated under the same experimental conditions using the original knowledge passages and the corresponding question-aware summarized passages. This allows us to assess whether the benefits of summarization generalize across different hallucination detection approaches.

TABLE III: Verification model training hyperparameters.

Parameter	Value
Backbone	DeBERTa-v3-base
Max sequence length	512
Optimizer	AdamW
Learning rate	2×10^{-5}
Weight decay	0.01
Warmup ratio	0.05
Train batch size	8
Eval batch size	8
Epochs	2
Split	70/30 stratified

C. Hyperparameter Selection

The hyperparameters in Table III were selected based on commonly adopted transformer fine-tuning practices and preliminary validation experiments on the training split. The learning rate of 2×10^{-5} was chosen because it provided stable optimization without large validation fluctuations. The batch size was set to 8 due to GPU memory limitations when using a maximum sequence length of 512 tokens. A warmup ratio of 0.05 and weight decay of 0.01 were used to improve training stability and reduce overfitting. The model was trained for two epochs because validation performance converged rapidly, as shown in Table IV, and additional training was not necessary.

For reproducibility, all experiments were conducted using the HuggingFace Transformers framework on an NVIDIA Tesla T4 GPU. A fixed random seed of 42 was used for data shuffling, train-test splitting, and model initialization. All reported results were obtained using the same 70/30 stratified split and the same hyperparameter configuration.The verifier was trained for two epochs, requiring approximately 1.5 hours of total training time.

D. Training Convergence Analysis

To analyze convergence behavior, we report the training and validation metrics across epochs. As shown in Table IV, the verifier improves from epoch 1 to epoch 2, with validation loss decreasing and F1 increasing. These results indicate that the model converges rapidly within two epochs while maintaining stable validation performance.

TABLE IV: Training convergence of the DeBERTa verifier.

Epoch	Training Loss	Validation Loss	Accuracy	Precision	F1
1	0.4414	0.4057	0.9760	0.9612	0.9764
2	0.4183	0.3940	0.9770	0.9640	0.9773

E. Stability Across Random Seeds

To evaluate the robustness and reproducibility of the proposed DeBERTa-based verifier, we repeated the complete training and evaluation procedure on the question-aware FLAN-T5 summarized knowledge setting using three different random seeds (42, 123, and 999) while keeping all hyperparameters, data splits, and training settings unchanged.

TABLE V: Performance stability across three random seeds on the FLAN-T5 summarized knowledge setting.

Seed	Accuracy	Precision	Recall	F1
42	0.9848	0.9815	0.9883	0.9849
123	0.9787	0.9674	0.9907	0.9789
999	0.9787	0.9656	0.9927	0.9790
Mean	0.9807	0.9715	0.9906	0.9809
Std	0.0035	0.0087	0.0021	0.0035

Table V reports the performance obtained for each seed on the FLAN-T5 summarized knowledge setting, together with the mean and standard deviation across runs. The results show only minor variations in accuracy, precision, recall, and F1 score, indicating that the proposed verifier is highly stable and does not depend on a particular random initialization.

The main FLAN-T5 summarized result reported in Table VII corresponds to the run with random seed 42. The additional runs with seeds 123 and 999 demonstrate that the reported performance remains stable across different random initializations.

The average F1 score across the three runs is 0.9809 with a standard deviation of only 0.0035, demonstrating strong reproducibility and consistent optimization behavior. Furthermore, the mean F1 score remains close to the main result reported in Table VII (0.9849), with a difference of only 0.0040. These results provide additional evidence that the reported performance is reliable and not attributable to seed-specific effects.

F. Zero-shot NLI Baseline

To isolate the contribution of task-specific fine-tuning, we evaluate a zero-shot NLI DeBERTa baseline on the same 6,000-sample HaluEval-QA test split. The input is formulated as a natural language inference problem, where the knowledge passage and question are treated as the premise, and the candidate answer is treated as the hypothesis. The entailment label is mapped to supported answer prediction, while neutral and contradiction labels are mapped to hallucination.

As shown in Table VI, the zero-shot NLI baseline achieves only 0.5280 accuracy and 0.2559 F1. This is substantially lower than the fine-tuned verifier, indicating that the strong performance of the proposed approach is not simply due to using a pretrained DeBERTa backbone. Instead, the results show that supervised task-specific fine-tuning is essential for reliable evidence-answer verification in HaluEval-QA.

TABLE VI: Comparison between zero-shot NLI and task-specific fine-tuned verification on the HaluEval-QA test set.

Method	Accuracy	Precision	Recall	F1
Zero-shot NLI DeBERTa	0.5280	0.6042	0.1623	0.2559
Fine-tuned DeBERTa	0.9770	0.9640	0.9910	0.9773
Ours + FLAN-T5	**0.9848**	**0.9815**	**0.9883**	**0.9849**

G. Main Results

Table VII compares verifier performance using original knowledge passages and the best summarization setting observed in our experiments. Using summarized knowledge improves accuracy and F1 while reducing input length.

TABLE VII: Impact of different summarization strategies on downstream hallucination detection.

Summarization Strategy	Accuracy	Precision	Recall	F1
Original (No summarization)	0.9770	0.9640	0.9910	0.9773
Overall Summarization (Generic)	0.9767	0.9631	0.9913	0.9770
Question-Aware BART-CNN	0.9790	0.9666	0.9923	0.9793
Question-Aware PEGASUS	0.9782	0.9656	0.9917	0.9785
Question-Aware FLAN-T5	**0.9848**	**0.9815**	**0.9883**	**0.9849**

H. Example Verification Outputs

To further illustrate the behavior of the proposed verifier, the following examples show two predictions generated by the model on the HaluEval-QA test set.

Example 1 (Supported Answer Correctly Verified by the Proposed Model)
Question: Hanni Toosbuy Kasprzak is a Danish billionaire and the owner of what Danish shoe manufacturer and retailer founded in 1963 in Bredebro, Denmark?
Answer: ECCO
Ground Truth: Supported
Model Prediction: Supported
Interpretation: The verifier correctly identifies that the answer is explicitly supported by the provided evidence and therefore classifies it as a non-hallucinated response.

Example 2 (Hallucinated Answer Correctly Detected by the Proposed Model)

Question: What Pulitzer prize winning, American composer known for exploring addiction and homophobia, wrote "Tick, Tick...Boom!"?
Answer: "Tick, Tick...Boom!" was written by Stephen Sondheim.
Ground Truth: Hallucinated
Model Prediction: Hallucinated
Interpretation: The verifier correctly detects that the answer introduces information that is not supported by the evidence and therefore classifies it as a hallucinated response.

I. Summarization Faithfulness Validation

To validate the quality of the generated summaries, we perform a dedicated faithfulness evaluation using MiniCheck. The original knowledge passage is treated as the source document, while the generated summary is treated as the claim. The resulting faithfulness score reaches 92.98%, indicating that most summaries remain supported by the original evidence.

We additionally measure explicit answer preservation and find that 74.78% of summaries explicitly retain the reference answer. Despite reducing the average context length by 61.05%, the summaries maintain a high level of factual consistency, supporting their use as a preprocessing step for hallucination detection.

Table VIII summarizes the results.

TABLE VIII: Faithfulness validation of the FLAN-T5 summaries used in the proposed framework.

Metric	Value
Summary Faithfulness (MiniCheck)	92.98%
Answer Preservation Rate	74.78%
Compression Ratio	0.389
Context Reduction	61.05%

The difference between summary faithfulness and answer preservation is expected because the two metrics measure different properties. Summary faithfulness evaluates whether the generated summary is factually supported by the original knowledge passage, whereas answer preservation measures whether the reference answer is explicitly retained in the summary. Therefore, a summary can remain faithful to the original evidence while omitting the exact answer span or expressing the relevant evidence implicitly. This explains why the faithfulness score is higher than the explicit answer preservation rate.

J. Statistical Significance

To verify that the observed performance gain is not due to random variation, we conduct McNemar's test on paired predictions from the original and summarized settings. McNemar's test confirms that the improvement obtained using question-aware FLAN-T5 summarization is statistically significant ($\chi^2 = 21.37$, exact $p = 2.48 \times 10^{-6}$), indicating consistent gains across samples.

K. Ablation Study

To better understand the contribution of each component, we conduct an ablation study using different knowledge processing strategies while keeping the verifier architecture unchanged.

TABLE IX: Ablation study of the proposed framework on HaluEval-QA.

Configuration	Accuracy	F1
DeBERTa Verifier Only	0.9770	0.9773
+ Generic Summarization	0.9767	0.9770
+ Question-Aware BART	0.9790	0.9793
+ Question-Aware PEGASUS	0.9782	0.9785
+ Question-Aware FLAN-T5	**0.9848**	**0.9849**

Table IX shows that the baseline DeBERTa verifier already achieves strong performance using the original knowledge passages. Generic summarization does not provide a meaningful improvement, indicating that compression alone is insufficient.

In contrast, question-aware summarization consistently improves verification performance across all evaluated summarization backbones. Among them, FLAN-T5 achieves the best results, improving the F1 score from 0.9773 to 0.9849.

These results demonstrate that the primary gain comes from providing the verifier with question-focused evidence rather than simply reducing context length.

L. Comparison with Existing Hallucination Detection Baselines

Table X compares the proposed verifier against a zero-shot NLI baseline and two widely used factual consistency evaluation frameworks, namely AlignScore and MiniCheck.

TABLE X: Comparison with external hallucination detection baselines on the HaluEval-QA test set.

Method	Accuracy	Precision	Recall	F1
Zero-shot NLI DeBERTa	0.5280	0.6042	0.1623	0.2559
Lexical Exact Match (Original)	0.8527	0.9962	0.7080	0.8277
Lexical Exact Match (Summarized)	0.8822	0.9948	0.7683	0.8670
MiniCheck [11] (Original)	0.7378	0.7381	0.7378	0.7377
MiniCheck [11] (Summarized)	0.7400	0.7487	0.7400	0.7377
AlignScore [10] (Original)	0.7708	0.7227	0.8790	0.7932
AlignScore [10] (Summarized)	0.7928	0.7777	0.8198	0.7982
Ours (Original)	0.9770	0.9640	0.9910	0.9773
Ours + FLAN-T5	**0.9848**	**0.9815**	**0.9883**	**0.9849**

We additionally evaluate a simple lexical exact-match baseline that predicts a supported answer when the candidate answer appears verbatim in the knowledge passage. The lexical baseline achieves an F1 score of 0.8277 on the original data and 0.8670 on the summarized data, indicating that lexical overlap provides a strong signal in HaluEval-QA. However, the proposed DeBERTa verifier substantially outperforms the lexical baseline, suggesting that successful hallucination detection requires semantic evidence-answer verification beyond surface-level lexical matching.

The zero-shot NLI baseline performs substantially worse than all task-specific methods, achieving only 0.2559 F1. This result highlights the importance of supervised fine-tuning for evidence-answer verification and demonstrates that the gains achieved by the proposed approach are not solely due to the underlying pretrained DeBERTa backbone.

Among the evaluated baselines, AlignScore achieves the strongest performance, reaching an F1 score of 0.7982 when applied to question-aware summarized knowledge. MiniCheck achieves an F1 score of approximately 0.74 under both original and summarized settings.

In contrast, the proposed DeBERTa-based verifier substantially outperforms both baselines, achieving an F1 score of 0.9773 using the original knowledge passages and 0.9849 when combined with FLAN-T5 question-aware summarization.

The results indicate that direct supervised optimization on (K, Q, A) triples is considerably more effective for hallucination detection than generic factual consistency scoring methods. Furthermore, all evaluated methods benefit from summarized knowledge, suggesting that reducing irrelevant context improves evidence-answer alignment across different verification frameworks.

M. Evaluation Across Transformer Backbones

To further evaluate the effectiveness of the proposed framework, we compare DeBERTa-v3 with several widely used transformer verification backbones, including BERT, RoBERTa, and ELECTRA. All models were fine-tuned and evaluated under the same experimental protocol using question-aware FLAN-T5 summarized knowledge.

TABLE XI: Comparison of fine-tuned transformer verifiers using question-aware FLAN-T5 summarized knowledge.

Model	Accuracy	Precision	Recall	F1
BERT-base	0.9322	0.9602	0.9017	0.9300
ELECTRA-base	0.9412	0.9607	0.9200	0.9399
RoBERTa-base	0.9625	0.9636	0.9613	0.9625
DeBERTa-v3-base (Ours)	**0.9848**	**0.9815**	**0.9883**	**0.9849**

As shown in Table XI, all transformer verifiers achieve strong performance on the hallucination detection task. However, the proposed DeBERTa-v3 verifier consistently outperforms the alternative backbones, achieving the highest accuracy (0.9848) and F1 score (0.9849). RoBERTa achieves the second-best performance with an F1 score of 0.9625, followed by ELECTRA (0.9399) and BERT (0.9300).

These results demonstrate that the proposed framework is not limited to a specific backbone architecture. Nevertheless, the combination of question-aware FLAN-T5 summarization and DeBERTa-v3 provides the strongest evidence-answer verification capability among all evaluated transformer models.

N. Analysis and Visualization

Figure 2 shows that summarization substantially reduces knowledge length. Figure 3 presents the confusion matrices for original and summarized knowledge. Summarization mainly reduces false positives. Table VII summarizes the overall metric improvements across summarization strategies.

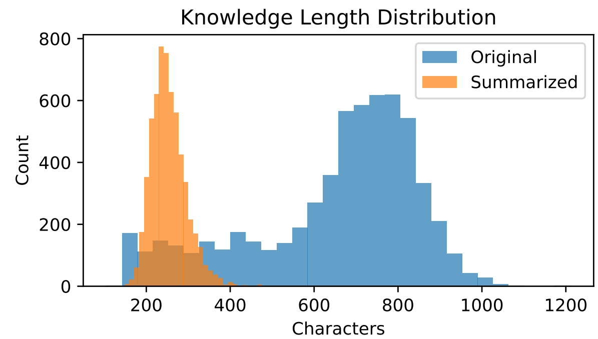
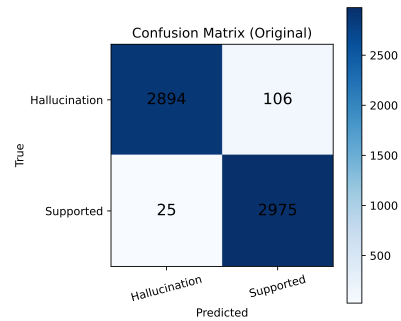
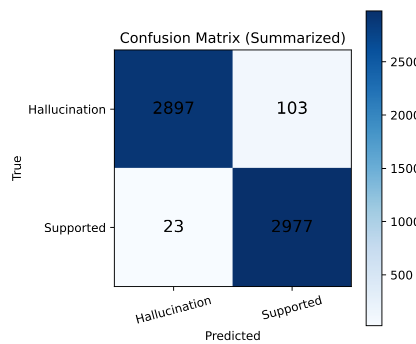


Fig. 2: Knowledge length distribution before and after summarization.



(a) Original knowledge



(b) Summarized knowledge

Fig. 3: Confusion matrices using original and summarized knowledge passages on the HaluEval-QA dataset.

O. Inference Efficiency Analysis

To quantify the computational benefits of question-aware knowledge summarization, we measured inference efficiency on the 6,000-sample HaluEval-QA test set using the same fine-tuned DeBERTa-v3 verifier and identical hardware settings. The evaluation compares original knowledge passages against their summarized counterparts.

TABLE XII: Inference efficiency comparison between original and summarized knowledge.

Setting	Avg Tokens	Latency (ms)	Throughput (samples/s)	GPU Time (s)	GPU Mem. (GB)
Original	110.95	15.99	62.56	95.91	2.68
Summarized	**63.25**	**9.71**	**103.02**	**58.24**	**1.59**

As shown in Table XII, question-aware summarization reduces the average input length from 110.95 to 63.25 tokens. This reduction decreases inference latency by approximately 39% and reduces total GPU processing time from 95.91 s to 58.24 s. Furthermore, throughput increases from 62.56 to 103.02 samples/s, while peak GPU memory consumption decreases by approximately 41%.These results provide direct empirical evidence that question-aware summarization reduces computational cost while simultaneously improving hallucination detection performance.

P. Error Analysis

To better understand the remaining failure cases, we analyzed the 138 misclassified samples from the HaluEval-QA test set, corresponding to an error rate of 2.3%. The errors were grouped into coarse-grained categories using a heuristic keyword-based inspection of the question, answer, and evidence fields.

TABLE XIII: Categorization of remaining verification errors on the HaluEval-QA test set.

Error Category	Count	Percentage
Temporal Reasoning	32	23.19%
Location Confusion	29	21.01%
Entity Confusion	25	18.12%
Multi-hop / Comparative Reasoning	21	15.22%
Subtle Evidence Mismatch	19	13.77%
Numerical Reasoning	7	5.07%
Lexical Ambiguity	5	3.62%

As shown in Table XIII, most remaining errors are associated with temporal reasoning, location-related confusion, and entity disambiguation. These categories require the verifier to distinguish fine-grained factual details such as dates, places, and named entities. Multi-hop and comparative questions also contribute to a noticeable portion of the errors, suggesting that some failures occur when the answer depends on aggregating evidence across multiple facts. In contrast, numerical reasoning and lexical ambiguity account for a smaller portion of the residual errors.

Overall, the analysis indicates that the proposed verifier is highly effective for direct evidence-answer matching, while the remaining errors are concentrated in cases requiring fine-grained reasoning, entity disambiguation, or subtle evidence comparison.

1) Representative Error Cases: To better understand the remaining errors, we present representative examples from three of the most frequent failure categories identified in Table XIII.

Temporal Reasoning Failure. Temporal reasoning errors occur when the verifier struggles to distinguish between closely related dates, years, or event timelines. Although the candidate answer may appear plausible, it is not fully supported by the evidence due to a subtle temporal mismatch.

Example: The verifier incorrectly accepted the answer “March 1994” for a question asking about the album release date associated with The Prodigy’s single *Poison*. This example illustrates the difficulty of verifying fine-grained temporal information when multiple dates appear in the evidence.

Entity Confusion Failure. Entity confusion occurs when multiple people, organizations, or locations appear within the same context and the verifier incorrectly associates an attribute with the wrong entity. Such errors are common when entities share similar roles or appear together in the evidence.

Example: For a question asking at which university the scientist who co-authored *The Quantum Universe: Everything That Can Happen Does Happen* is a professor, the verifier incorrectly rejected the supported answer “University of Manchester”. The error arose from difficulty identifying the correct entity among multiple referenced individuals.

Multi-hop Reasoning Failure. Multi-hop reasoning errors occur when determining answer correctness requires combining multiple pieces of evidence rather than relying on a single supporting statement. The verifier may correctly recognize individual facts but fail to integrate them into a complete reasoning chain.

Example: A representative failure involved determining which album was produced by John Leckie, engineered by Nigel Godrich, and featured artwork by the same artist responsible for another album by the band. The verifier incorrectly accepted the answer “OK Computer”, demonstrating the challenge of aggregating evidence across multiple related facts.

Q. Generalization on a HotpotQA-Derived Binary Benchmark

To further evaluate the generalization capability of our approach, we conduct an additional external experiment using a binary hallucination-detection benchmark derived from HotpotQA. HotpotQA was not used during training, which allows us to assess the behavior of the verifier under an unseen distribution with longer and more reasoning-intensive contexts.

TABLE XIV: Generalization results on the HotpotQA-derived binary hallucination-detection benchmark.

Setting	Total	Accuracy	Macro Precision	Macro F1
Lexical Exact Match (Original)	1978	0.6571	0.9670	0.4868
Lexical Exact Match (Summarized)	1978	0.7768	0.9507	0.7234
Original knowledge	1978	0.5025	0.5035	0.4662
Question-aware FLAN-T5 summarization	1978	0.7917	0.7988	0.7905

Unlike the original HotpotQA question-answering format, we convert the data into a balanced binary verification setting aligned with our task formulation. Each instance is represented as a triple (K, Q, A) and assigned a binary label indicating whether the candidate answer is supported by the provided knowledge passage or hallucinated.

For each original HotpotQA example, the ground-truth answer is used to create a supported sample (label = 1). Hallucinated samples (label = 0) are generated using a type-aware negative sampling strategy. Specifically, the original answer is replaced with an answer sampled from a different HotpotQA instance while preserving the answer type whenever possible (e.g., person, location, date, number, or yes/no). Additional filtering is applied to remove duplicate answer pairs and trivial replacements.

The resulting benchmark contains 1,978 samples, with 989 supported answers and 989 hallucinated answers. This balanced setup avoids class-prior bias and provides a fairer comparison between the original and summarized knowledge settings.

HotpotQA contexts are typically long and multi-hop, often containing substantial irrelevant information. Such noise can make evidence-answer verification difficult, especially when the answer depends on a small subset of the provided passage. Therefore, we evaluate whether question-aware summarization improves verification by reducing irrelevant context while preserving the evidence needed to judge the candidate answer.

We evaluate the same verification pipeline under two settings: using the original knowledge passages and using question-aware summarized passages. The summarized passages are generated using the same FLAN-T5 question-aware summarization procedure described in Section IV. As shown in Table XIV, the FLAN-T5 summarization setting achieves the best performance. Without summarization, the verifier obtains an accuracy of 0.5025 and a macro-F1 score of 0.4662, which is close to random performance on this balanced binary benchmark.

After applying question-aware summarization, performance improves substantially to 0.7917 accuracy and 0.7905 macro-F1. This corresponds to an absolute improvement of approximately 29 percentage points in accuracy and more than 32 percentage points in macro-F1.

The lexical baseline's strong showing on HaluEval-QA (0.8277 F1) is telling. We read it not as a real strength of the heuristic but as a sign of lexical bias in the dataset: a hallucinated answer can often be spotted by surface overlap alone, with no reasoning involved. HotpotQA looks very different. There the same baseline falls to 0.4868 F1 on the original passages — essentially chance — and the verifier on those same passages drops to 0.4662 macro-F1. Summarization does lift the lexical baseline (to 0.7234), simply because a shorter passage brings the answer closer to the surface and makes matching easier; but the verifier gains more, and on cases the lexical method still misses, which suggests summarization sharpens the evidence rather than just shortening it. We therefore treat HaluEval-QA as a controlled in-distribution check and HotpotQA as the harder test of whether the model is really reasoning over the evidence.

R. Discussion

Unlike retrieval-augmented, ensemble-based, and judge-based approaches, the proposed framework achieves competitive performance using a single deterministic verifier and a lightweight preprocessing stage. This demonstrates that strong hallucination detection performance can be obtained without architectural modifications or complex multi-stage pipelines.

Summarization significantly improves hallucination detection performance while reducing input length. The improvement is statistically significant and achieved without modifying the verifier architecture, preserving simplicity and deployability.

The baseline comparison further highlights the effectiveness of the proposed approach. While AlignScore and MiniCheck provide useful reference-based consistency signals, both methods remain substantially below the performance achieved by the supervised DeBERTa verifier. The strongest baseline, AlignScore, reaches an F1 score of 0.7982, whereas the proposed model achieves an F1 score of 0.9849 under the FLAN-T5 summarization setting.

Interestingly, all evaluated methods benefit from summarized knowledge, including both reference-based consistency metrics and fine-tuned transformer verifiers. This suggests that reducing irrelevant context improves evidence-answer consistency assessment independently of the underlying verification model.

Furthermore, the superiority of the summarized setting is maintained across multiple transformer verification architectures, suggesting that the benefit of question-aware evidence compression is not limited to a specific backbone.

However, the much larger improvement observed for the proposed verifier indicates that task-specific supervised optimization remains more effective than generic factual consistency scoring approaches for hallucination detection in knowledge-grounded QA.

VII. Practical Deployment Considerations

The proposed framework is designed for real-world deployment. Unlike white-box approaches, it does not require access to internal model states. Unlike ensemble-based methods, it relies on a single lightweight verifier, reducing inference cost and engineering overhead.

As summarized in Table I, the proposed approach combines black-box compatibility, deterministic inference, and low system complexity, making it suitable for practical deployment scenarios. Unlike retrieval-augmented, ensemble-based, or judge-based frameworks, the proposed verifier operates as a single end-to-end model and therefore avoids additional engineering overhead and dependence on external components.

The summarization step further improves efficiency by reducing input length. In our setting, summarized passages are substantially shorter than the original knowledge, which can translate into lower encoding latency and higher throughput at inference time. While exact speedups depend on hardware and batch size, shorter inputs generally reduce encoder computation and memory usage, making the approach attractive under strict latency constraints.

In addition to improving efficiency, question-aware summarization consistently improves detection performance across both HaluEval-QA and the external HotpotQA-derived benchmark. This indicates that the framework can benefit from reduced context length without sacrificing verification quality, making it suitable for long-context QA applications.

In practice, the system can be deployed as a plug-and-play verification module: generate (or retrieve) knowledge, optionally summarize it conditioned on the question, and then verify candidate answers. Because the verifier operates directly on textual inputs and does not require access to model internals, it can be integrated into existing black-box QA systems with minimal modification.

Overall, the combination of a lightweight supervised verifier and optional question-aware summarization provides a practical balance between performance, efficiency, reproducibility, and deployment simplicity.

VIII. Conclusion

We presented an efficient supervised hallucination detection framework for knowledge-grounded QA by fine-tuning DeBERTa-v3-base as a verifier over (K, Q, A) triples. Despite its simplicity, the verifier achieves strong performance on HaluEval-QA while maintaining competitive generalization on the external HotpotQA-derived benchmark, with statistically significant improvements when combined with question-aware knowledge summarization. Across multiple summarization backbones, FLAN-T5 delivered the best downstream detection results in our experiments.

Our findings demonstrate that reducing irrelevant context is a key factor for improving hallucination detection. Question-aware summarization consistently improves evidence-answer alignment, reduces noise in long knowledge passages, and enhances both verification performance and external generalization. A lightweight supervised verifier provides an effective mechanism for exploiting these improved evidence representations without relying on multi-stage retrieval pipelines or sampling-based evaluation strategies.

Overall, the proposed pipeline is simple, reproducible, cost-effective, and suitable for deployment in real-world QA systems, achieving strong performance on HaluEval-QA while maintaining competitive generalization on the external HotpotQA-derived benchmark.

Additional comparisons with AlignScore and MiniCheck further validate the effectiveness of the proposed verifier. Although both baselines benefit from question-aware summarization, the proposed approach consistently achieves substantially higher accuracy and F1 scores on the HaluEval-QA benchmark.

Future work can extend this framework to more challenging settings, including noisy or multi-document retrieval, distribution shifts, and adaptive calibration or abstention strategies to further improve robustness.

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